



PIMPRI CHINCHWAD EDUCATION TRUST'S.
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
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Affiliated to Savitribai Phule Pune University (SPPU)
ISO 21001:2018 Certified by TUV SUD

T.Y. B. TECH

Year: 2025 – 26

Subject: FSDL

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Branch: Computer Engineering

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Assignment No 7

Title : Design a web application for e-commerce or students feedback review systems or any other applications with MongoDB Backend. using React.js and Express.

Objectives:

- To create an E-commerce system or Student Feedback Review system
- To use MongoDB for storing user data, product details, and feedback information
- To understand frontend and backend integration
- To build and use REST APIs for communication between client and server
- To provide a simple and user-friendly interface for users

Outcomes:

- Learn how to connect and use MongoDB database
- Develop REST APIs to handle user requests
- Perform CRUD operations (Create, Read, Update, Delete)
- Gain knowledge of real-world web application development
- Improve understanding of client-server architecture

Theory :

- The Student Feedback Review System is a web-based application developed using React.js, Node.js, Express.js, and MongoDB.
- This system allows students to:
 - Submit feedback about teachers or subjects
 - Give ratings and comments
 - View previously submitted feedback (if allowed)
- The main aim of this system is to collect student opinions and improve teaching quality.
- Express.js: A lightweight framework of Node.js used to create APIs and handle routing and server requests easily.
- MongoDB: A NoSQL database that stores data in flexible JSON-like format, allowing easy scalability and fast access.

Architecture (Working):

This application follows a client-server architecture, where:

- The client (React.js frontend) provides a user-friendly interface for students
- The server (Node.js + Express.js) handles requests and processes data
- The database (MongoDB) stores feedback, student details, and ratings

Technologies Used:**• React.js:**

Used to create interactive and dynamic user interface. It helps in building reusable components.

• Node.js:

Used for backend development. It handles multiple requests efficiently using non-blocking operations.

• Express.js:

Used to create APIs and manage routing between frontend and backend.

• MongoDB:

A NoSQL database used to store feedback data in flexible JSON format.

Working of the System:

- Students enter their details and submit feedback through the form
- React frontend sends the data to backend using API requests
- Express server receives the data and processes it
- MongoDB stores the feedback information
- Admin or system can fetch and display feedback when required

Key Features:

- Easy feedback submission form
- Rating and comment system
- Data stored securely in database
- Fast and responsive UI using React
- CRUD operations (Create, Read, Update, Delete)

Code :

Server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
app.use(cors());
app.use(express.json());

// Updated Connection - Using a fresh DB name 'collegeFeedback'
mongoose.connect("mongodb+srv://admin:12345@admin.wcpc0dk.mongodb.net/collegeFeedback")
  .then(() => console.log("MongoDB Connected ✅"))
  .catch(err => console.log("DB Error:", err));

// Expanded Schema
const FeedbackSchema = new mongoose.Schema({
  name: String,
  subject: String,
  facultyName: String,
  rating: Number,
  comment: String,
  date: { type: Date, default: Date.now }
});

const Feedback = mongoose.model("Feedback", FeedbackSchema);

// GET All
app.get("/feedback", async (req, res) => {
  try {
    const data = await Feedback.find().sort({ date: -1 });
    res.json(data);
  } catch (err) {
    res.status(500).json({ error: "Failed to fetch" });
  }
});

// POST New
app.post("/feedback", async (req, res) => {
  try {
    const newData = new Feedback(req.body);
    await newData.save();
    res.json(newData);
  } catch (err) {
    res.status(500).json({ error: "Error saving feedback" });
  }
});
```

```
}  
});
```

```
app.listen(5000, () => console.log("Server running on port 5000 🚀"));
```

index.html

```
<!DOCTYPE html>  
<html lang="en">  
  //manasvipatil vala mongo  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Academic Feedback Portal</title>  
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0/css/all.min.css">  
  <style>  
    :root {  
      --primary: #2c3e50;  
      --accent: #3498db;  
      --success: #27ae60;  
      --light: #ecf0f1;  
    }  
  
    body {  
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;  
      background: #f0f2f5;  
      margin: 0;  
      color: var(--primary);  
    }  
  
    /* Navbar */  
    .navbar {  
      background: var(--primary);  
      color: white;  
      padding: 1rem 2rem;  
      display: flex;  
      justify-content: space-between;  
      align-items: center;  
      box-shadow: 0 2px 5px rgba(0,0,0,0.1);  
    }  
  
    .container {  
      max-width: 800px;  
      margin: 40px auto;  
      background: white;
```

```
padding: 30px;
border-radius: 12px;
box-shadow: 0 4px 15px rgba(0,0,0,0.05);
}
```

```
h2 { border-bottom: 2px solid var(--accent); padding-bottom: 10px; }
```

```
/* Form Styling */
```

```
.form-group { margin-bottom: 15px; text-align: left; }
label { display: block; font-weight: bold; margin-bottom: 5px; }
```

```
input, select, textarea {
  width: 100%;
  padding: 12px;
  border: 1px solid #ddd;
  border-radius: 6px;
  box-sizing: border-box;
}
```

```
/* Star Rating */
```

```
.stars {
  display: flex;
  flex-direction: row-reverse;
  justify-content: flex-end;
  gap: 10px;
  margin: 10px 0;
}
.stars input { display: none; }
.stars label {
  font-size: 30px;
  color: #ccc;
  cursor: pointer;
  transition: 0.2s;
}
.stars input:checked ~ label,
.stars label:hover,
.stars label:hover ~ label { color: #f1c40f; }
```

```
button {
  background: var(--accent);
  color: white;
  border: none;
  padding: 12px 25px;
  border-radius: 6px;
  cursor: pointer;
  font-size: 16px;
  font-weight: bold;
  transition: 0.3s;
}
```

```

    }
    button:hover { background: #2980b9; }

    /* Table Styling */
    #adminView { display: none; margin-top: 30px; }
    table {
        width: 100%;
        border-collapse: collapse;
        margin-top: 20px;
        background: white;
    }
    th, td {
        padding: 12px;
        border: 1px solid #ddd;
        text-align: left;
    }
    th { background: var(--primary); color: white; }
    tr:nth-child(even) { background: #f9f9f9; }

    /* Login Modal */
    #loginOverlay {
        display: none;
        position: fixed;
        top:0; left:0; width:100%; height:100%;
        background: rgba(0,0,0,0.8);
        justify-content: center;
        align-items: center;
        z-index: 1000;
    }
    .login-box {
        background: white;
        padding: 30px;
        border-radius: 8px;
        width: 300px;
    }
</style>
</head>
<body>

<div class="navbar">
    <div style="font-size: 1.5rem; font-weight: bold;">EduFeedback</div>
    <button onclick="showLogin()" style="background: rgba(255,255,255,0.2);">Admin Login</button>
</div>

<div class="container" id="formView">
    <h2>Student Feedback Form</h2>

    <div class="form-group">

```

```
<label>Your Name</label>
<input id="name" placeholder="Enter your full name">
</div>
```

```
<div class="form-group">
  <label>Subject</label>
  <select id="subject">
    <option value="">Select Subject</option>
    <option value="Data Structures">Data Structures</option>
    <option value="Operating Systems">Operating Systems</option>
    <option value="Artificial Intelligence">Artificial Intelligence</option>
    <option value="Computer Networks">Computer Networks</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>Faculty Name</label>
  <select id="faculty">
    <option value="">Select Teacher</option>
    <option value="Dr. Smith">Dr. Smith</option>
    <option value="Prof. Johnson">Prof. Ganesh Patil</option>
    <option value="Dr. Williams">Dr. Pratik</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>Rating</label>
  <div class="stars">
    <input type="radio" name="star" id="star5" value="5"><label for="star5">★</label>
    <input type="radio" name="star" id="star4" value="4"><label for="star4">★</label>
    <input type="radio" name="star" id="star3" value="3"><label for="star3">★</label>
    <input type="radio" name="star" id="star2" value="2"><label for="star2">★</label>
    <input type="radio" name="star" id="star1" value="1"><label for="star1">★</label>
  </div>
</div>
```

```
<div class="form-group">
  <label>Comments</label>
  <textarea id="comment" rows="3" placeholder="Tell us more..."></textarea>
</div>
```

```
<button onclick="submitFeedback()" style="width: 100%;">Submit Feedback</button>
</div>
```

```
<div id="loginOverlay">
  <div class="login-box">
    <h3>Admin Access</h3>
```

```

        <input type="text" id="adminUser" placeholder="Username" style="margin-bottom:10px;">
        <input type="password" id="adminPass" placeholder="Password" style="margin-bottom:20px;">
        <button onclick="verifyLogin()" style="width: 100%;">Login</button>
        <button onclick="closeLogin()" style="background: #e74c3c; margin-top:10px;
width:100%;">Cancel</button>
    </div>
</div>

<div class="container" id="adminView">
    <div style="display: flex; justify-content: space-between; align-items: center;">
        <h2>All Feedback Records</h2>
        <button onclick="window.location.reload()" style="background: var(--success);">Logout</button>
    </div>
    <table>
        <thead>
            <tr>
                <th>Student</th>
                <th>Subject</th>
                <th>Faculty</th>
                <th>Rating</th>
                <th>Comment</th>
            </tr>
        </thead>
        <tbody id="tableBody"></tbody>
    </table>
</div>

<script>
    // Star Rating Helper
    function getSelectedRating() {
        const stars = document.getElementsByName('star');
        for (let star of stars) {
            if (star.checked) return star.value;
        }
        return 0;
    }

    async function submitFeedback() {
        const rating = getSelectedRating();
        const data = {
            name: document.getElementById("name").value,
            subject: document.getElementById("subject").value,
            facultyName: document.getElementById("faculty").value,
            rating: rating,
            comment: document.getElementById("comment").value
        };

        if (!data.name || !data.subject || rating == 0) {

```



```

        alert("Please fill all required fields! ⚠️");
        return;
    }

    try {
        await fetch("http://localhost:5000/feedback", {
            method: "POST",
            headers: { "Content-Type": "application/json" },
            body: JSON.stringify(data)
        });
        alert("Feedback submitted successfully! ✨");
        location.reload();
    } catch (err) {
        alert("Server Error ❌");
    }
}

// Admin Logic
function showLogin() {
    document.getElementById("loginOverlay").style.display = "flex";
}

function closeLogin() {
    document.getElementById("loginOverlay").style.display = "none";
}

function verifyLogin() {
    const user = document.getElementById("adminUser").value;
    const pass = document.getElementById("adminPass").value;

    if (user === "admin" && pass === "admin@123") {
        document.getElementById("loginOverlay").style.display = "none";
        document.getElementById("formView").style.display = "none";
        document.getElementById("adminView").style.display = "block";
        loadFeedback();
    } else {
        alert("Invalid Credentials! 🚫");
    }
}

async function loadFeedback() {
    const res = await fetch("http://localhost:5000/feedback");
    const data = await res.json();
    const tbody = document.getElementById("tableBody");
    tbody.innerHTML = "";

    data.forEach(f => {

```

```

const row = `<tr>
  <td>${f.name}</td>
  <td>${f.subject}</td>
  <td>${f.facultyName}</td>
  <td style="color: #f1c40f; font-weight:bold;">${f.rating} ★</td>
  <td>${f.comment}</td>
</tr>`;
tbody.innerHTML += row;
});
}
</script>

</body>
</html>

```

Output :

EduFeedback
Admin Login

Student Feedback Form

Your Name

Subject

Faculty Name

Rating

★ ★ ★ ★ ★

Comments

Submit Feedback

Student Feedback Form

Your Name

Enter your full name

Subject

Select Subject

Faculty Name

Select Teacher

Rating



Comments

Tell us more...

Submit Feedback

Admin Access

admin

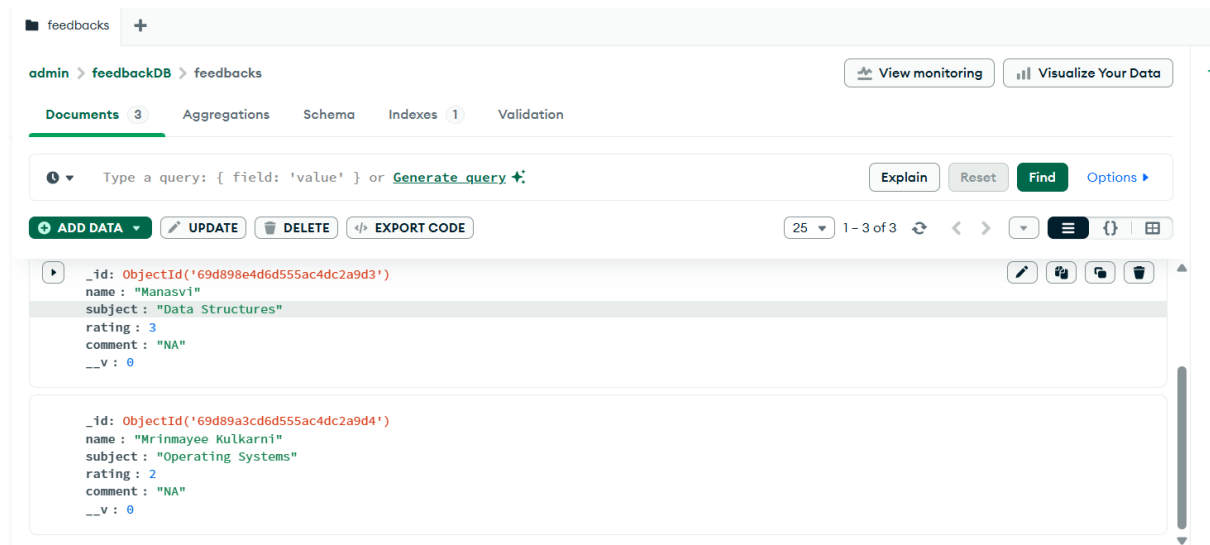
Login

Cancel

All Feedback Records

Logout

Student	Subject	Faculty	Rating	Comment
Manasvi Patil	Math	undefined	3 ★	1
Manasvi	Data Structures	undefined	3 ★	NA
Mrinmayee Kulkarni	Operating Systems	undefined	2 ★	NA
Tanisha	Artificial Intelligence	undefined	3 ★	Not Applicable



Conclusion :

The Student Feedback Review System developed using React.js, Node.js, Express, and MongoDB provides a simple and effective way to collect, store, and manage student feedback. It demonstrates the integration of frontend and backend technologies along with database management. This system helps in understanding full-stack development and improves the process of analyzing feedback for better decision-making and enhancement of educational quality.